

Remote Sensing

Rhett D Harrison

Introduction

To do

Fire incidence

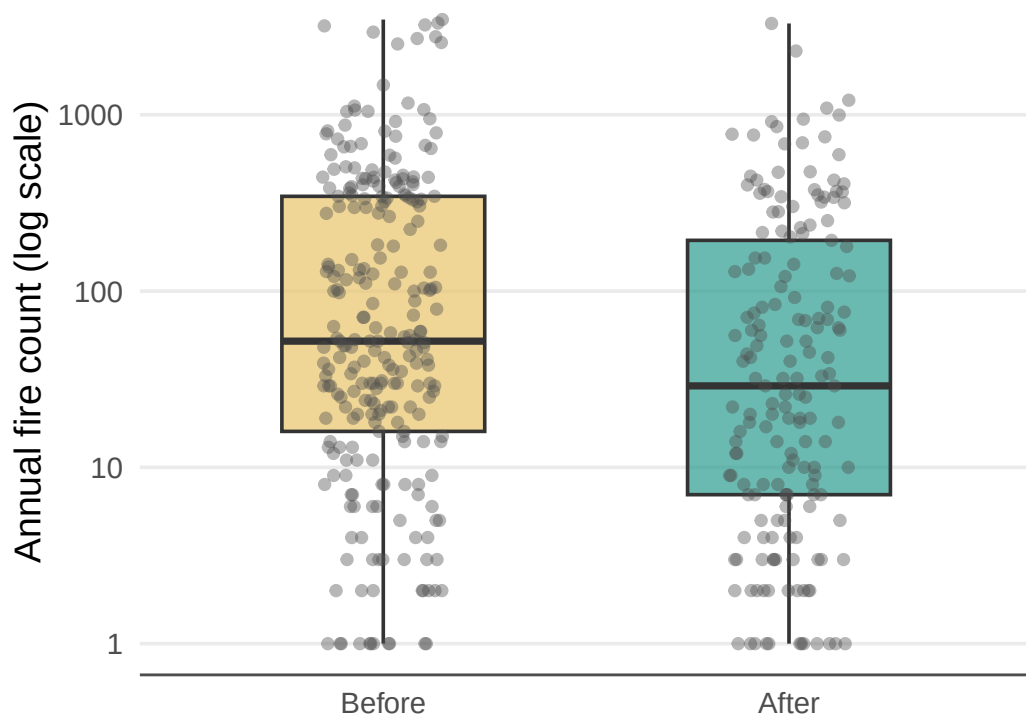


Figure 1: Annual fire detection count per CFMG before and after establishment. Each point represents one CFMG-year. Boxes show median and interquartile range; y-axis is on a log scale.

Table 1: Linear mixed model results for log-transformed annual fire count before and after CF establishment.

Model results for annual fire count (log-transformed) — before vs after establishment
 Linear mixed model with random intercept for CFMG. Reference = Before. CFMG variance = 3.779; residual variance = 0.21. Marginal $R^2 = 0.001$; conditional $R^2 = 0.947$. $n = 423$.

Factor	Estimate	SE	<i>P</i> value
Intercept (Before)	3.788	0.309	0.0000
After	−0.126	0.048	0.0091

Burned area

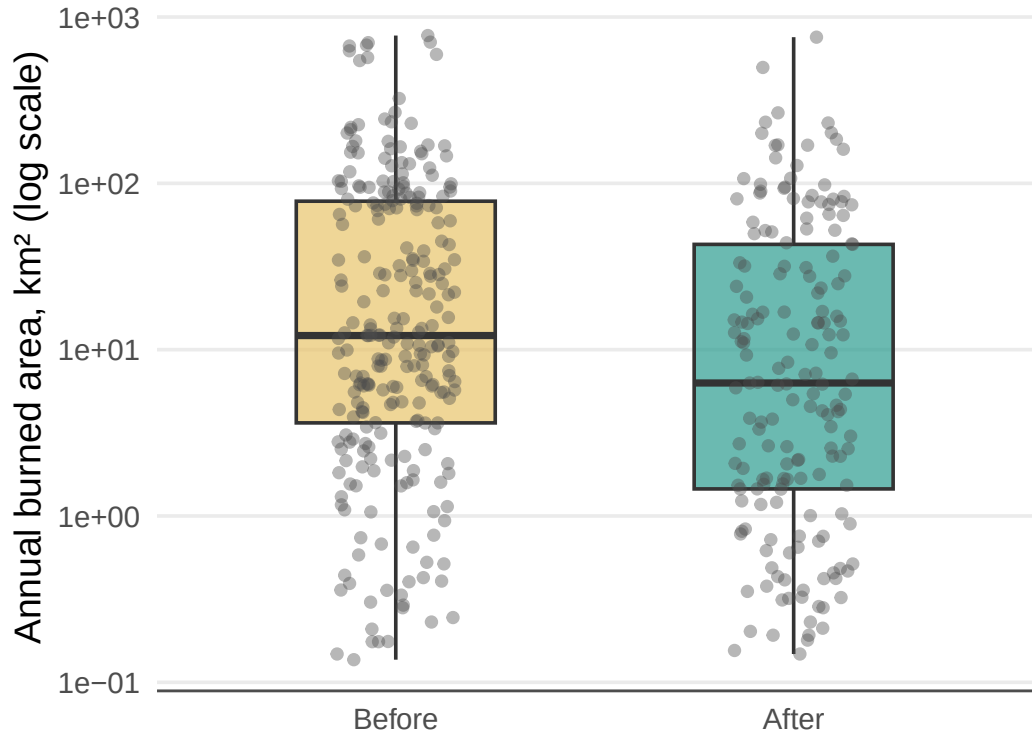


Figure 2: Annual burned area (km²) per CFMG before and after establishment. Burned area estimated as the sum of scan $\times$ track for all fire detections within each CFMG-year. Each point represents one CFMG-year. Boxes show median and interquartile range; y-axis is on a log scale.

Table 2: Linear mixed model results for log-transformed annual burned area before and after CF establishment.

Model results for annual burned area (log-transformed) — before vs after establishment
 Linear mixed model with random intercept for CFMG. Reference = Before. CFMG variance = 2.893; residual variance = 0.148. Marginal $R^2 = 0.001$; conditional $R^2 = 0.951$. $n = 423$.

Factor	Estimate	SE	<i>P</i> value
Intercept (Before)	2.538	0.270	0.0000
After	−0.135	0.040	0.0009

Fire radiative power

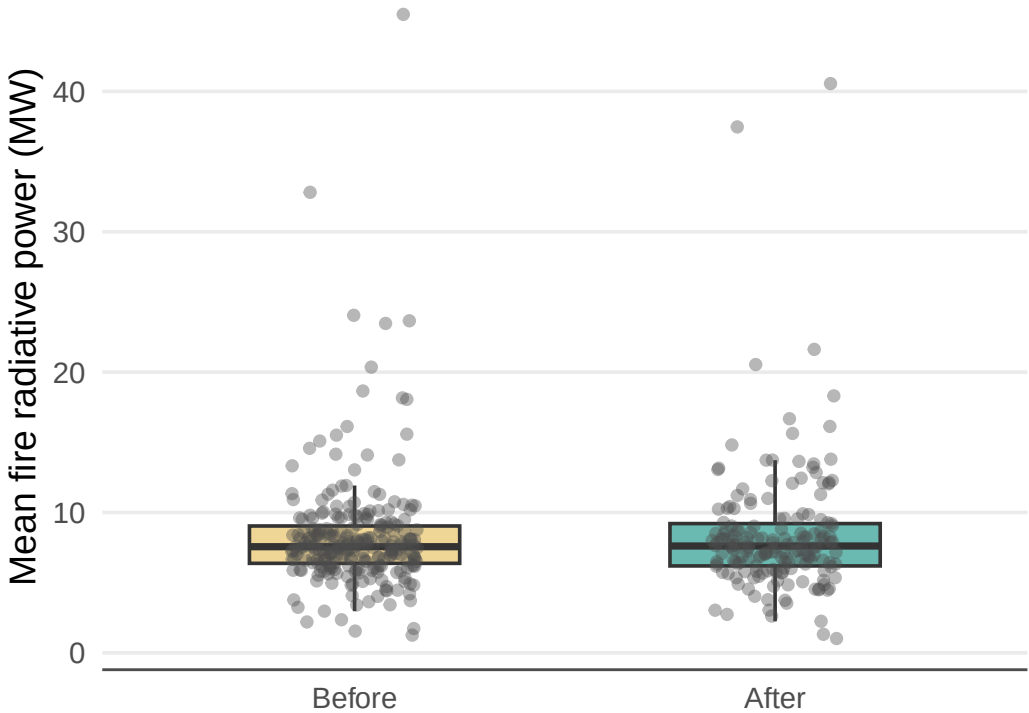


Figure 3: Mean annual fire radiative power (FRP, MW) per CFMG before and after establishment. FRP is averaged across all fire detections within each CFMG-year. Each point represents one CFMG-year. Boxes show median and interquartile range.

Table 3: Linear mixed model results for mean annual fire radiative power before and after CF establishment.

Model results for mean annual FRP — before vs after establishment

Linear mixed model with random intercept for CFMG. Reference = Before. CFMG variance = 6.103; residual variance = 14.758. Marginal $R^2 = 0.001$; conditional $R^2 = 0.293$. $n = 423$.

Factor	Estimate	SE	P value
Intercept (Before)	8.472	0.465	0.0000
After	−0.312	0.399	0.4346

Deforestation

To do

Discussion

To do